

Ravi Singh

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PROFESSIONAL SUMMARY

Senior Java Backend Engineer and Tech Lead with **9+ years** in fintech, core banking, and PCI-DSS payments security. At Osfin, architected a tokenization platform processing **50M+ card transactions** (PCI-DSS L1), re-engineered the reconciliation engine for a **7.5x speedup (30 min → 4 min on 10M records)**, and delivered **10M-card key rotation in <1 minute** across OCI, AWS, GCP, and Azure. **Led agentic AI adoption** - built a Claude-powered platform for PR review, PRD-to-code, test/security checks, and RCAs, driving **30-40% faster delivery**. Tech Owner for 2-3 banking clients with full production accountability.

CORE COMPETENCIES

Java / Spring Boot

PCI-DSS L1 Tokenization

Multi-Cloud KMS (OCI/AWS/GCP/Azure)

Event-Driven Architecture

Agentic AI / Claude Code

RabbitMQ · AWS SQS

Kubernetes · Docker

Tech Lead / Tech Owner

WORK EXPERIENCE

Osfin April 2023 - Present

Senior Software Engineer (Tech Lead) · Bengaluru, India

Leading a **6-engineer cross-functional team**; Tech Owner for 2-3 banking clients (financial reconciliation platform, 170+ data connectors) with full production SLA, deployments, and migrations.

PCI-DSS Tokenization & Encryption Platform (independently architected)

- Architected** an end-to-end tokenization engine processing **50M+ card transactions** at sub-millisecond latency under PCI-DSS L1 - replacing raw PANs across all payment touchpoints.
- Delivered** cryptographic key rotation for **10M cards in <1 minute** via a high-concurrency pipeline with zero customer impact.
- Engineered** cross-cloud key management across **OCI KMS, AWS KMS, GCP Cloud KMS, and Azure Key Vault** with zero-downtime key lifecycle ops.
- Integrated** HashiCorp Vault for dynamic secrets and encryption-as-a-service; AES-256/RSA pipelines with **zero critical vulnerabilities** across all audits.
- Shipped** zero-downtime blue-green deployments on OKE/EKS/GKE via Jenkins and GitHub Actions; mentored team on PCI-DSS and secure coding.

Agentic AI Engineering Platform - Claude-powered SDLC Automation (led and architected)

- Architected and rolled out** an internal agentic-AI platform on Claude Code automating **PR review (~120 PRs/wk), PRD-to-code, test generation, security checks, and RCA drafts** - **~6 hrs/engineer/week recovered, PR-merge time down 35%, and ~40% of security findings caught pre-merge**, measured via cycle-time analytics across the 6-engineer team.
- Built custom plugins and skills** to automate recurring engineering tasks (weekly releases, documentation generation, changelogs).
- Authored** PR-review and security guardrails enforcing PCI-DSS coding standards; drove team-wide adoption via prompt libraries and Claude Code workflows.

Reconciliation & Payments Event Pipeline

- Led re-architecture** of the recon engine from batch SQL to in-memory streaming across a **6-engineer team** - **30 min → 4 min on 10M records (7.5x speedup)**, **memory footprint cut 60%**, and surfaced via **OpenTelemetry traces** that cut P1 MTTR from **45 → 18 min**.
- Built** async event ingestion across **RabbitMQ, AWS SQS, OCI Queue, Azure Service Bus** - millions of multi-bank events daily with **idempotent consumers**, ordered delivery, **Resilience4j circuit breakers**, and DLQ recovery.
- Achieved 99.5%+ auto-match accuracy**; reduced manual reconciliation effort by **70%**; **<5-minute downtime** on schema/infra migrations with P1 RCA ownership.

Stack: Java 11/17, Spring Boot/Security/Batch, Resilience4j, Oracle, MySQL, Redis, RabbitMQ, AWS SQS, OCI Queue, Azure Service Bus, HashiCorp Vault, OCI/AWS/GCP/Azure KMS, AES-256, RSA, OAuth 2.0, JWT, PCI-DSS, Docker, Kubernetes, GitOps, Jenkins, GitHub Actions, JUnit 5, Mockito, Testcontainers, SonarQube, OpenTelemetry, Claude / Claude Code, GitHub Copilot.

Java Full Stack Developer · Pune, India

Bee-Recon (client: Games24x7) - Enterprise reconciliation across FX accounts, card terminals, and bank transfers

- Engineered reconciliation engine processing **300K+ daily transactions** across 8+ payment channels with sub-second match latency.
- Improved batch throughput by **45%** via parallel Spring Batch pipelines; cut integration failures by **60%** with retry + DLQ logic across 5+ banking APIs.
- Implemented Spring Security RBAC and structured logging - cut incident resolution by **35%**.

Core Banking System (client: Sanjivani Credit Cooperative)

- Built Spring + Hibernate + Angular 4 stack covering accounts, deposits, loans, and transaction workflows.

Stack: Java 7/8, J2EE, Spring 3, Hibernate, REST, JUnit, Tomcat, MySQL, Angular 4/5, Log4j2, Maven.

EDUCATION

Bachelor of Engineering - Electronics and Communication Apex Institute of Engineering & Technology (AIET), Jaipur

2017

SKILLS

Languages: Java 8/11/17, SQL, TypeScript, JavaScript

Backend & APIs: Spring Boot, Spring Security, Spring Batch, Hibernate, REST, gRPC, GraphQL, OpenAPI, Microservices, Event-Driven Architecture

Architecture & Resilience: CQRS, Event Sourcing, Saga, Outbox, Idempotency, Circuit Breaker (Resilience4j), Retry, Bulkhead, Rate Limiting, Backpressure, JVM tuning

Cloud: OCI (OKE, KMS, Queue) · AWS (EKS, S3, SQS, Secrets Manager) · GCP (GKE, Cloud KMS) · Azure (AKS, Key Vault, Service Bus)

Messaging: RabbitMQ, AWS SQS, OCI Queue, Azure Service Bus Databases: Oracle, MySQL, SQL Server, Redis

Security: PCI-DSS L1, Tokenization, AES-256, RSA, HSM-backed KMS, HashiCorp Vault, OAuth 2.0, JWT, Zero-Trust

Testing & Quality: JUnit 5, Mockito, Testcontainers, Pact, WireMock, SonarQube, code coverage

DevOps & Observability: Docker, Kubernetes, Jenkins, GitHub Actions, GitOps, Prometheus, Grafana, ELK, OpenTelemetry, Distributed Tracing

Agentic AI: Claude / Claude Code, Custom Plugins & Skills, Agentic Workflows (PR review, PRD-to-code, test/security automation, RCAs), GitHub Copilot

Leadership: Architecture Reviews, Threat Modeling, P1 Incident Response, Mentoring